Esraa Adel Shehatta Ali

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OBJECTIVE

AI & Machine Learning Engineer Dedicated technologist with a strong foundation in Deep Learning and NLP. Experienced in developing intelligent systems using LLMs and Generative AI frameworks. Expertise includes implementing RAG architectures for improved factual accuracy in AI responses, data preprocessing for structured/unstructured sources, and training robust models across Computer Vision and Robotics domains.

EDUCATION

Bachelor of Computer Science (B.SC) |Artificial Intelligence Specialization
Banha University — 2025
GPA: 2.95/4.0

Professional Trainings

Global Brands Group |New Cairo, Egypt
AI& Generative AI Diploma | February 2026 – Present

- Specializing in Generative AI architectures, including the development and fine-tuning of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) systems.
- Building end-to-end Generative AI applications using LangChain, Hugging Face, and vector databases to handle complex unstructured data.
- Implementing advanced deep learning techniques and prompt engineering to optimize model performance for domain-specific tasks.
- Ongoing Project: Developing scalable AI solutions that leverage state-of-the-art Generative models for real-world business automation.

National Telecommunication Institute(NTI) | Cairo, Egypt
Digital Twin Engineer Intern (HireReady Program) | October 2025 – January 2026

- Engineered a Digital Twin for an ICU Decision Management System, integrating real-time IoT data with AI-driven predictive modeling to optimize patient triage and hospital resources.
- Developed autonomous navigation systems for robotics using A* and BFS pathfinding algorithms within the Webots simulation environment.
- Implemented communication protocols (MQTT/HTTP) to bridge physical sensor data with digital representations, ensuring low-latency data synchronization.
- Collaborated on multi-disciplinary tasks involving Unity/Unreal Engine and Python to visualize complex data analytics in a 3D environment.

National Telecommunication Institute (NTI) | Qalyoubia, Egypt
Machine Learning Intern (Creativa Innovation Hubs) | April 2025 – June 2025

- Applied advanced supervised and unsupervised learning algorithms to solve data-driven challenges and predictive modeling tasks.
- Developed and deployed a Heart Failure Prediction system using Python and Streamlit, achieving high accuracy in clinical risk assessment.
- Managed the full data lifecycle, from cleaning and feature engineering to model evaluation and deployment.

PROJECTS

Heart Failure Predictive Analytics System

- Architected an end-to-end machine learning pipeline to assess cardiovascular risk, achieving a peak accuracy of 98% using the UCI Clinical Records dataset.
- Engineered a comparative study of ensemble methods, including Random Forest and XGBoost, optimizing performance through rigorous feature selection and hyperparameter tuning.
- Developed and launched a real-time clinical decision-support tool using Streamlit, enabling interactive patient data input and immediate risk visualization.
- Tech Stack: Python, Scikit-learn, XGBoost, Pandas, Matplotlib, Streamlit.

Neural Machine Translation (NMT) Engine

- Built a high-performance English-to-Spanish translation engine leveraging Transformer architectures and Sequence-to-Sequence (Seq2Seq) modeling.
- Integrated attention mechanisms to improve context retention and linguistic accuracy across long-form translations.
- Preprocessed large-scale linguistic datasets, implementing tokenization and padding strategies to optimize training efficiency.
- Tech Stack: Python, TensorFlow, NLP, Transformers, Seq2Seq.

Graduation Project - Robot Localization

- Implemented an AI-driven system to determine robotic navigation and positioning.
- Utilized neural networks to process real-time data for precise location pinpointing.
- Integrated GPS data to improve localization accuracy under varying environmental conditions.
- Key Technologies: Python, TensorFlow, Neural Networks, GPS Data Integration.

Natural Language to SQL (NL2SQL) Database Agent

- Developed a sophisticated AI Chatbot capable of querying PostgreSQL databases using natural language through LLM-driven SQL generation.
- Implemented a RAG-style workflow where the system interprets user intent, generates schema-aware SQL queries, and translates raw data results back into human-readable insights.
- Built a responsive user interface using Streamlit and integrated the Gemini API for high-accuracy reasoning and query optimization.
- Optimized prompt engineering strategies to ensure secure and accurate database interactions without manual SQL coding.
- Tech Stack: Python, Generative AI (LLMs), LangChain, PostgreSQL, Streamlit, Gemini API.

TECHNICAL SKILLS

- Programming & Scripting: Python, C++, HTML, JavaScript
- Machine Learning & AI: Supervised/Unsupervised Learning, Neural Networks, Transformers, Predictive Modeling, NLP, Model Evaluation & Optimization
- Libraries & Frameworks: NumPy, Pandas, TensorFlow, PyTorch , Scikit-learn, OpenCV, NLTK, Matplotlib,
- Seaborn, PySpark, , LangChain, Hugging Face, crawl4ai.
- Data Management: MySQL, SQL, Data Cleaning, Feature Engineering, Data Preprocessing Data Visualization & BI Tools: Power BI, Tableau, Excel
- Other Tools: Git, Github
- Generative AI**:** LLMs (GPT-4, Llama), Prompt Engineering, RAG (Retrieval-Augmented Generation).
- Soft Skills: Analytical Thinking, Problem-Solving, Adaptability, Collaboration, Continuous Learning.

CERTIFICATIONS

- Data Engineering (AWS Academy)
- Machine Learning for Data Analysis (NTI)
- Career Development level 1-2-3 (NTI)
- Getting Started with Deep Learning (NVIDIA)
- Introduction to Transformer-Based Natural Language Processing (NVIDIA)